

Investing Money - Simple Interest - Repeated Use

Simple interest investments pay their interest at the end of the investment. So if the investment is for 3 years, the interest is paid at the end of that 3 year period. Compound interest investments pay interest during the investment. For example a 3 year investment might have interest credited after the first, second and third years.

This doesn't necessarily mean that you get your interest earlier. The interest is invested with the original principal. So this means that for the second year the principal is higher with an increase in the amount of interest earned. The third year will again see an increase in the Principal with an increase in interest.

So compound investments earn more interest, this sheet deals with calculating the final interest payment without using the Compound Interest formula, which is introduced in the next outcome.

The method is simple, you apply the simple interest formula multiple times. There is a short cut however and this is shown on Sheet 2. Here is an example.

Example: An investment of \$4 000 at 6% p.a., is compounded annually for 3 years.

Find the account balance, A , at the end of the investment.

First Application: Year 1

$$I = Prn$$

$$= 4\,000 \times 0.06 \times 1$$

$$I = \$240$$

$$A = 4\,000 + 240$$

$$A = \$4\,240$$

Second Application: Year 2

$$I = Prn$$

$$= 4\,240 \times 0.06 \times 1$$

$$I = \$254.40$$

$$A = 4\,240 + 254.40$$

$$A = \$4\,494.40$$

Third Application: Year 3

$$I = Prn$$

$$= 4\,494.40 \times 0.06 \times 1$$

$$I = \$269.66$$

$$A = 4\,494.40 + 269.66$$

$$A = \$4\,764.06$$

Note how the principal increases each year. Instead of being \$4 000 for three years, it increases each year. So the interest earned also increases.

As you can see, instead of earning \$240 every year, each year sees an increase in the amount of interest earned. So compound investments offer a better return than simple interest investments. As long as they are for a longer duration than 1 compound period.

Investing Money - Simple Interest - Repeated Use (Sheet 1)

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When an investment compounds, interest is paid during the investment. Calculate the account balance for these.

Principal \$2 000,
Rate = 7% p.a.,
 $n = 3$ yrs, compounding annually.

First Application: Year 1

1 $I = Prn$

$$= \boxed{} \times \boxed{} \times 1$$

Principal Rate (decimal)

$I = \boxed{}$ Show '\$' sign

$A = \boxed{} + \boxed{}$

Principal Interest (Q1)

$A = \boxed{}$

Second Application: Year 2

2 $I = Prn$

$$= \boxed{} \times \boxed{} \times 1$$

Principal (Use A-Yr 1) Rate (decimal)

$I = \boxed{}$

$A = \boxed{} + \boxed{}$

Principal (Use A-Yr 1) Interest (Q2)

$A = \boxed{}$

Third (Final) Application: Year 3

3 $I = Prn$

$$= \boxed{} \times \boxed{} \times 1$$

Principal (Use A-Yr 2) Rate (decimal)

$I = \boxed{}$

$A = \boxed{} + \boxed{}$

Principal (Use A-Yr 2) Interest (Q3)

$A = \boxed{}$

Compare the compounding investment (Q 1-3) to a standard simple interest investment.

4 $I = Prn$

$$= \boxed{} \times \boxed{} \times 3$$

\$2 000 Rate 7% p.a.

$I = \boxed{}$

$A = \boxed{} + \boxed{}$

Principal Interest (Q4)

$A = \boxed{}$

5 Find the difference between your answer for 'A' in Q3 & Q4

Diff = $\boxed{} - \boxed{}$

A (Q3) A (Q4)

= $\boxed{}$

Now try a 2 year compounding investment. Find the account balance.

Principal \$920,
Rate = 8.4% p.a.,
 $n = 2$ yrs, compounding annually

First Application: Year 1

6 $I = Prn$

$$= \boxed{} \times \boxed{} \times 1$$

$I = \boxed{}$

$A = \boxed{} + \boxed{}$

$A = \boxed{}$

Second (Final) Application: Year 2

7 $I = Prn$

=

$I =$

$A =$

$A =$

Compare the compounding investment (Q 6-7) to a standard simple interest investment.

8 $I = Prn$

$$= \boxed{} \times \boxed{} \times 2$$

\$920 Rate 8.4% p.a.

$I = \boxed{}$

$A = \boxed{} + \boxed{}$

Principal Interest (Q8)

$A = \boxed{}$

9 Find the difference between your answer for 'A' in Q7 & Q8

Diff = $\boxed{} - \boxed{}$

A (Q7) A (Q8)

= $\boxed{}$

Now use a different approach to answering compound questions. Calculate these

10 Principal \$2 000,
Rate = 7% p.a., $[1 + 0.07 = 1.07]$
 $n = 3$ yrs, compounding annually.

$A = 2\,000 \times 1.07 \times 1.07 \times 1.07$

Principal Repeated 3 times as $n = 3$

$A = \boxed{}$

11 Is this the same as your answer in Q3? ☐ Yes ☐ No

12 Principal \$920,
Rate = 8.4% p.a., $[1 + r = 1.084]$
 $n = 2$ yrs, compounding annually

$A = 920 \times 1.084 \times 1.084$

Principal Repeated 2 times as $n = 2$

$A = \boxed{}$

13 Is this the same as your answer in Q7? ☐ Yes ☐ No

Investing Money - Simple Interest - Repeated Use (Sheet 2)

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Use multiplication to find the account balance more quickly. Change the rate to "1.xx", where 'xx' is the rate as a decimal. Find the account balance, A .



- 1 Principal \$5 000, Rate = 6% p.a., $n = 4$ yrs, compounding annually.

$$A = 5\,000 \times 1.06 \times 1.06 \times 1.06 \times 1.06$$

$$A = \boxed{}$$

- 2 Principal \$3 100, Rate = 9% p.a., $n = 5$ yrs, compounding annually.

$$A = 3\,100 \times 1.09 \times 1.09 \times 1.09 \times 1.09 \times 1.09$$

$$A = \boxed{}$$

- 3 Principal \$665, Rate = 5% p.a., $n = 4$ yrs, compounding annually.

$$A = \boxed{} \times \boxed{} \times \boxed{} \times \boxed{} \times \boxed{}$$

$$A = \boxed{}$$



Did you use 1.05?
Not 0.05.

- 4 Principal \$10 444, Rate = 12% p.a., $n = 3$ yrs, compounding annually.

$$A =$$

$$A =$$

- 5 Principal \$850.70, Rate = 8.2% p.a., $n = 2$ yrs, compounding annually.

$$A =$$

$$A =$$

- 6 Principal \$72 310, Rate = 6.7% p.a., $n = 4$ yrs, compounding annually.

$$A =$$

$$A =$$

- 7 Principal \$40 700, Rate = 11.6% p.a., $n = 3$ yrs, compounding annually.

$$A =$$

$$A =$$



Instead of using "1.xx"... n times, use powers to find A . Once you find A , find I by subtracting the principal from A .

- 8 Principal \$6 100, Rate = 4.8% p.a., $n = 7$ yrs, compounding annually.

$$A = 6\,100 \times 1.048^7 = \boxed{}$$

$$\text{Calc: } \boxed{6100} \times \boxed{1.048}^{\boxed{7}} =$$

$$I = \boxed{} - \boxed{} = \boxed{}$$

- 9 Principal \$7 040, Rate = 6.6% p.a., $n = 4$ yrs, compounding annually.

$$A = 7\,040 \times 1.066^4 = \boxed{}$$

$$I = \boxed{} - \boxed{} = \boxed{}$$

- 10 Principal \$370, Rate = 9% p.a., $n = 2$ yrs, compounding annually.

$$A = \boxed{} \times 1.\boxed{}\boxed{} = \boxed{}$$

$$I = \boxed{} - \boxed{} = \boxed{}$$

- 11 Principal \$24 112, Rate = 7.3% p.a., $n = 8$ yrs, compounding annually.

$$A = \boxed{} \times \boxed{}^{\boxed{}} = \boxed{}$$

$$I = \boxed{} - \boxed{} = \boxed{}$$

- 12 Principal \$1 770, Rate = 6.2% p.a., $n = 5$ yrs, compounding annually.

$$A = \boxed{} \times \boxed{}^{\boxed{}} = \boxed{}$$

$$I = \boxed{} - \boxed{} = \boxed{}$$

Yu invests \$5 000 at 8.3% p.a. for 6 years, compounded annually. Find his account balance and interest earned.

13

$$A = \boxed{} \times \boxed{}^{\boxed{}} = \boxed{}$$

$$I = \boxed{} - \boxed{} = \boxed{}$$

Investing Money - Simple Interest - Repeated Use (Sheet 1)

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When an investment compounds, interest is paid during the investment. Calculate the account balance for these.

Principal \$2 000,
Rate = 7% p.a.,
 $n = 3$ yrs, compounding annually.

First Application: Year 1

$$1 \quad I = Prn$$

$$= \boxed{2\,000} \times \boxed{0.07} \times 1$$

Principal Rate (decimal)

$$I = \boxed{\$140}$$

Show '\$' sign

$$A = \boxed{2\,000} + \boxed{140}$$

Principal Interest (Q1)

$$A = \boxed{\$2\,140}$$

Second Application: Year 2

$$2 \quad I = Prn$$

$$= \boxed{2\,140} \times \boxed{0.07} \times 1$$

Principal (Use A-Yr 1) Rate (decimal)

$$I = \boxed{\$149.80}$$

$$A = \boxed{2\,140} + \boxed{149.80}$$

Principal (Use A-Yr 1) Interest (Q2)

$$A = \boxed{\$2\,289.80}$$

Third (Final) Application: Year 3

$$3 \quad I = Prn$$

$$= \boxed{2\,289.80} \times \boxed{0.07} \times 1$$

Principal (Use A-Yr 2) Rate (decimal)

$$I = \boxed{\$160.29}$$

$$A = \boxed{2\,289.80} + \boxed{160.29}$$

Principal (Use A-Yr 2) Interest (Q3)

$$A = \boxed{\$2\,450.09}$$

Compare the compounding investment (Q 1-3) to a standard simple interest investment.

$$4 \quad I = Prn$$

$$= \boxed{2\,000} \times \boxed{0.07} \times 3$$

\$2 000 Rate 7% p.a.

$$I = \boxed{\$420}$$

$$A = \boxed{2\,000} + \boxed{420}$$

Principal Interest (Q4)

$$A = \boxed{\$2\,420}$$

5 Find the difference between your answer for 'A' in Q3 & Q4

$$\text{Diff} = \boxed{2\,450.09} - \boxed{2\,420}$$

$$= \boxed{\$30.09}$$

Now try a 2 year compounding investment. Find the account balance.

Principal \$920,
Rate = 8.4% p.a.,
 $n = 2$ yrs, compounding annually

First Application: Year 1

$$6 \quad I = Prn$$

$$= \boxed{920} \times \boxed{0.084} \times 1$$

$$I = \boxed{\$77.28}$$

$$A = \boxed{920} + \boxed{77.28}$$

$$A = \boxed{\$997.28}$$

Second (Final) Application: Year 2

$$7 \quad I = Prn$$

$$= 997.28 \times 0.084 \times 1$$

$$I = \boxed{\$83.77}$$

$$A = 997.28 + 83.77$$

$$A = \boxed{\$1\,081.05}$$

Compare the compounding investment (Q 6-7) to a standard simple interest investment.

$$8 \quad I = Prn$$

$$= \boxed{920} \times \boxed{0.084} \times 2$$

\$920 Rate 8.4% p.a.

$$I = \boxed{\$154.56}$$

$$A = \boxed{920} + \boxed{154.56}$$

Principal Interest (Q8)

$$A = \boxed{\$1\,074.56}$$

9 Find the difference between your answer for 'A' in Q7 & Q8

$$\text{Diff} = \boxed{1\,081.05} - \boxed{1\,074.56}$$

$$= \boxed{\$6.49}$$

Now use a different approach to answering compound questions. Calculate these

10 Principal \$2 000,
Rate = 7% p.a., $[1 + 0.07 = 1.07]$
 $n = 3$ yrs, compounding annually.

$$A = 2\,000 \times 1.07 \times 1.07 \times 1.07$$

Principal Repeated 3 times as $n = 3$

$$A = \boxed{\$2\,450.09}$$

11 Is this the same as your answer in Q3? ☒ Yes ☐ No

12 Principal \$920,
Rate = 8.4% p.a., $[1 + r = 1.084]$
 $n = 2$ yrs, compounding annually

$$A = 920 \times 1.084 \times 1.084$$

Principal Repeated 2 times as $n = 2$

$$A = \boxed{\$1\,081.05}$$

13 Is this the same as your answer in Q7? ☒ Yes ☐ No

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Use multiplication to find the account balance more quickly. Change the rate to “1.xx”, where ‘xx’ is the rate as a decimal. Find the account balance, A .



- 1 Principal \$5 000, Rate = 6% p.a., $n = 4$ yrs, compounding annually.

$$A = 5\,000 \times 1.06 \times 1.06 \times 1.06 \times 1.06$$

$$A = \$6\,312.38$$

- 2 Principal \$3 100, Rate = 9% p.a., $n = 5$ yrs, compounding annually.

$$A = 3\,100 \times 1.09 \times 1.09 \times 1.09 \times 1.09 \times 1.09$$

$$A = \$4\,769.73$$

- 3 Principal \$665, Rate = 5% p.a., $n = 4$ yrs, compounding annually.

$$A = 665 \times 1.05 \times 1.05 \times 1.05 \times 1.05$$

$$A = \$808.31$$



Did you use 1.05?
Not 0.05.

- 4 Principal \$10 444, Rate = 12% p.a., $n = 3$ yrs, compounding annually.

$$A = 10\,444 \times 1.12 \times 1.12 \times 1.12$$

$$A = \$14\,673.07$$

- 5 Principal \$850.70, Rate = 8.2% p.a., $n = 2$ yrs, compounding annually.

$$A = 850.70 \times 1.082 \times 1.082$$

$$A = \$995.93$$

- 6 Principal \$72 310, Rate = 6.7% p.a., $n = 4$ yrs, compounding annually.

$$A = 72\,310 \times 1.067 \times 1.067 \times 1.067 \times 1.067$$

$$A = \$93\,725.13$$

- 7 Principal \$40 700, Rate = 11.6% p.a., $n = 3$ yrs, compounding annually.

$$A = 40\,700 \times 1.116 \times 1.116 \times 1.116$$

$$A = \$56\,570.11$$



Instead of using “1.xx”... n times, use powers of to find A . Once you find A , find I by subtracting the principal from A .

- 8 Principal \$6 100, Rate = 4.8% p.a., $n = 7$ yrs, compounding annually.

$$A = 6\,100 \times 1.048^7 = \$8\,469.52$$

$$\text{Calc: } 6100 \times 1.048^7 =$$

$$I = 8\,469.52 - 6\,100 = \$2\,369.52$$

- 9 Principal \$7 040, Rate = 6.6% p.a., $n = 4$ yrs, compounding annually.

$$A = 7\,040 \times 1.066^4 = \$9\,090.79$$

$$I = 9\,090.79 - 7\,040 = \$2\,050.79$$

- 10 Principal \$370, Rate = 9% p.a., $n = 2$ yrs, compounding annually.

$$A = 370 \times 1.09^2 = \$439.60$$

$$I = 439.60 - 370 = \$69.60$$

- 11 Principal \$24 112, Rate = 7.3% p.a., $n = 8$ yrs, compounding annually.

$$A = 24\,112 \times 1.073^8 = \$42\,367.32$$

$$I = 42\,367.32 - 24\,112 = \$18\,255.32$$

- 12 Principal \$1 770, Rate = 6.2% p.a., $n = 5$ yrs, compounding annually.

$$A = 1\,770 \times 1.062^5 = \$2\,391.09$$

$$I = 2\,391.09 - 1\,770 = \$621.09$$

Yu invests \$5 000 at 8.3% p.a. for 6 years, compounded annually. Find his account balance and interest earned.

13

$$A = 5\,000 \times 1.083^6 = \$8\,067.53$$

$$I = 8\,067.53 - 5\,000 = \$3\,067.53$$